

Qizhen Lan

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Houston, TX, United States

RESEARCH INTERESTS

- My research asks how AI can move from strong benchmark results to systems people can actually use. I work on efficient vision models and agentic AI, especially model compression and adaptation, multimodal reasoning, and workflow-oriented agents. Medical imaging is a key application of this work, where efficiency, reliability, and integration with clinical workflows are essential.

EDUCATION

- **University of Alabama at Birmingham** Jan 2024 – May 2025
Ph.D. in Computer Science Birmingham, United States
- **Bowling Green State University** Aug 2019 – Dec 2023
Ph.D. student in Data Science Bowling Green, United States
- **Bowling Green State University** Aug 2018 – Jul 2019
Master of Science in Analytics Bowling Green, United States
- **Bowling Green State University** Aug 2014 – Jun 2018
Bachelor of Science in Business Analytics and Intelligence Bowling Green, United States

PROFESSIONAL EXPERIENCE

- **University of Texas Health Science Center at Houston**  May 2025 – present
Postdoctoral Research Fellow (mentor: Dr. Xiaqian Jiang) Houston, TX, United States
 - Pretrained **3D brain foundation models** on large-scale MRI cohorts for Alzheimer's-related downstream tasks.
 - Built a **fast whole-brain segmentation pipeline** with approximately **~1s inference**, reducing processing time from FreeSurfer-scale hours to deployment-level latency.
 - Designed region- and context-aware distillation methods for compact 3D segmentation models, achieving **sub-10M-parameter** students while preserving downstream performance.
 - **Outcomes:** Three **NeurIPS 2026** papers accepted: BrainTRACE and OpenBrain in the Evaluations & Datasets Track, and ValuSpec as a Poster. One **npj Health Systems** paper accepted and three **MICCAI 2026** papers accepted, including **two early accepts**, with **one Spotlight and one Oral presentation**; last and corresponding author on two distillation papers; one paper nominated and shortlisted for the **Best Paper and Young Scientist Award**.
- **University of Alabama at Birmingham (UAB)**  Jan 2024 – May 2025
Research Assistant (Advisor: Dr. Qing Tian) Birmingham, AL, United States
 - Led optimization and deployment of object detection models on NVIDIA embedded GPUs (e.g., Jetson Xavier and Orin) for real-time autonomous-driving perception.
 - Developed adaptive **knowledge distillation** methods for dense prediction and DETR-style detectors, reducing model parameters by up to **70%** while maintaining detection performance.
 - **Outcomes:** First-author papers in **ICCV 2025**, **WACV 2026**, and **WACV 2024**, with additional publications in WACV, ICASSP, IV, and T-IV.
- **Bowling Green State University (BGSU)**  Aug 2021 – Jan 2024
Research Assistant (Advisor: Dr. Qing Tian) Bowling Green, OH, United States
 - Developed **efficient AI** algorithms for object detection models under autonomous-driving deployment constraints.
 - Built multi-factor and selective dark-knowledge distillation frameworks that reduce model size and computation while preserving detection accuracy across modern detectors.
 - **Outcomes:** IEEE T-IV journal paper and ICPR/WACV publications in efficient object detection and autonomous-driving perception.

- [M.1] **Qizhen Lan**, Yu-Chun Hsu, Nida Saddaf Khan, Xiaoqian Jiang. *Toward Clinically Deployable 3D Auto-Contouring: A Region- and Context-Aware Knowledge Distillation Framework*. **npj Digital Medicine**, 2026 (In preparation).
- [J.1] **Qizhen Lan**, Aaron Choi, Jun Ma, Bo Wang, Zhongming Zhao, Xiaoqian Jiang, Yu-Chun Hsu. *From Performance to Practice: Knowledge-Distilled Segmentator for On-Premises Clinical Workflows*. **npj Health Systems**, 2026.
- [J.2] Jung Im Choi*, **Qizhen Lan***, Qing Tian. *Boosting Deep Detector Efficiency and Robustness through Detection Discriminant Reorganization and Compression*. **Neural Networks**, 2026 (* Co-first author; Accepted).
- [J.3] **Qizhen Lan**, Qing Tian. *Instance, Scale, and Teacher Adaptive Knowledge Distillation for Visual Detection in Autonomous Driving*. **IEEE Transactions on Intelligent Vehicles (T-IV)**, 2022.
- [C.1] **Qizhen Lan**, Mengchen Fan, Hang Zhang, Jingwei Duan, Moule Lin, Jialin Chen, Baocheng Geng, and Xiaoqian Jiang. *BrainTRACE: Tracing Longitudinal, Multimodal, and Volumetric Evidence in Brain MRI Clinical Reasoning*. **NeurIPS Evaluations & Datasets Track**, 2026.
- [C.2] **Qizhen Lan**, Yu-Chun Hsu, Yuxiang Wei, Lijing Zhu, Zenan Sun, Liang He, Lishan Yu, and Xiaoqian Jiang. *OpenBrain: An Auditable Generated-Label Release for Whole-Brain MRI Parcellation*. **NeurIPS Evaluations & Datasets Track**, 2026.
- [C.3] Liang He, SiYuan Ma, Qishi Zhan, Yongqi Fan, Mingyu Cao, **Qizhen Lan**, Zhaolu Kang, and Xilu Wang. *ValuSpec: Plug-and-Play Candidate Valuation before Target Verification for Tree-Based Speculative Decoding*. **NeurIPS**, 2026 (Poster).
- [C.4] Jinwei Su*, **Qizhen Lan***, ZEYU WANG, Yinghui Xia, Hairu Wen, Yiqun Duan, Xi Xiao, TIANYU SHI, Yang Jingsong, and Lewei He. *ComfySearch: Autonomous Exploration and Reasoning for ComfyUI Workflows*. **EMNLP Findings**, 2026 (* Co-first author; Accepted). [Paper]
- [C.5] Tingyu Wu, CaoGuangyu, **Qizhen Lan**, Zhisheng Chen, Ziyang Weng, Bingkun Zhu, Miao Su, Chenglong Li, Zhengwei Xie, Huacan Wang, Sen Hu, Haocun Ye, NAN AN, Zaoqu Liu, and Ronghao Chen. *KnowMeBenchV2: Evidence-Grounded Person-Centric Long-Video Understanding*. **EMNLP Main Conference**, 2026 (Oral Presentation).
- [C.6] Zhisheng Chen, Yingwei Zhang, **Qizhen Lan**, Tianyu Liu, Huacan Wang, Yi Ding, Ziyu Jia, Ronghao Chen, Kun Wang, and Xinliang Zhou. *Uni-NTFM: A Unified Foundation Model for EEG Signal Representation Learning*. **ICLR**, 2026.
- [C.7] Jinwei Su*, **Qizhen Lan***, Yinghui Xia, Lifan Sun, Weiyou Tian, Tianyu Shi, Xinyuan Song, and Lewei He. *Difficulty-Aware Agentic Orchestration for Query-Specific Multi-Agent Workflows*. **WWW**, 2026 (* Co-first author).
- [C.8] Tingyu Wu, Zhisheng Chen, Ziyang Weng, Shuhe Wang, Chenglong Li, Shuo Zhang, Sen Hu, Silin Wu, **Qizhen Lan**[†], Huacan Wang, and Ronghao Chen. *KnowMe-Bench: Benchmarking Person Understanding for Lifelong Digital Companions*. **ACL Main Conference**, 2026 ([†] Corresponding author; Oral Presentation).
- [C.9] Xi Xiao, Chen Liu, Chih-Ting Liao, Yunbei Zhang, **Qizhen Lan**, Yuxiang Wei, Lin Zhao, Janet Wang, Jianyang Gu, Muchao Ye, Tianyang Wang, and Hao Xu. *Staying VIGILant: Mitigating Visual Laziness via Counterfactual Visual Alignment in MLLMs*. **ECCV**, 2026.
- [C.10] Mengchen Fan, Baocheng Geng, Xi Xiao, Tianyang Wang, Siyuan Mei, Pulin Che, Xiaoqian Jiang[†], and **Qizhen Lan**[†]. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **MICCAI**, 2026 ([†] Corresponding authors; Early Accept; Spotlight Presentation).
- [C.11] Zhicheng Ding*, Xinyu Chu*, Jung Im Choi, Qing Tian[†], Tianyu Shi, Xiaoqian Jiang, Lijing Zhu, and **Qizhen Lan**[†]. *Displacement Preserving Relational Distillation for Robust Medical Segmentation*. **MICCAI**, 2026 (* Co-first authors; [†] Corresponding authors). [Code]
- [C.12] Siyuan Mei, Yanteng Zhang, Yan Xia, **Qizhen Lan**, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Xiaoqian Jiang, Fuxin Fan, Yixing Huang, and Andreas Maier. *Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation*. **MICCAI**, 2026 (Early Accept; Oral Presentation; Nominated and Shortlisted for the Best Paper and Young Scientist Award).
- [C.13] **Qizhen Lan**, Qing Tian. *CLoCKDistill: Consistent Location-and-Context-aware Knowledge Distillation for DETRs*. **WACV**, 2026.
- [C.14] **Qizhen Lan**, Jong-Im Choi, Qing Tian. *Visual Detector Compression via Location-Aware Discriminant Analysis*. **WACV**, 2026.
- [C.15] Chen Liu*, **Qizhen Lan***, Zhicheng Ding, Xinyu Chu, Qing Tian. *Learnable Instance Attention Filtering for Adaptive Detector Distillation*. **ICASSP**, 2026 (* Co-first author).
- [C.16] **Qizhen Lan**, Qing Tian. *ACAM-KD: Adaptive and Cooperative Attention Masking for Knowledge Distillation*. **ICCV**, 2025.
- [C.17] Lijing Zhu, **Qizhen Lan**, Qing Tian, Wenbo Sun, Li Yang, Lu Xia, Yixin Xie, Xi Xiao, Tiehang Duan, Cui Tao, Shuteng Niu. *ETT-CKGE: Efficient Task-driven Tokens for Continual Knowledge Graph Embedding*. **ECML-PKDD**, 2025.
- [C.18] Yanteng Zhang, Songheng Li, Zeyu Shen, **Qizhen Lan**, Lipei Zhang, Yang Liu, Vince Calhoun. *Lost in Distortion: Uncovering the Domain Gap Between Computer Vision and Brain Imaging—A Study on Pretraining for Age Prediction*. **ISBI**, 2025.

[C.19]

Jong-Im Choi*, **Qizhen Lan***, Qing Tian. *Improving Deep Detector Robustness via Detection-Related Discriminant Maximization and Reorganization*. **WACV, 2025** (Oral; * Co-first author).

[C.20]

Zhicheng Ding, **Qizhen Lan**, Qing Tian. *Target-Driven and Student-Centered Knowledge Distillation for Traffic Object Tracking*. **IV, 2025**.

[C.21]

Qizhen Lan, Qing Tian. *Gradient-Guided Knowledge Distillation for Object Detectors*. **WACV, 2024**.

[C.22]

Wentao Wang, Xi Xiao, Mingjie Liu, **Qizhen Lan**, Xuanyao Huang, Qing Tian, Swalpa Kumar Roy, Tianyang Wang. *Multi-dimension Transformer with Attention-based Filtering for Medical Image Segmentation*. **ICTAI, 2024**.

HONORS AND AWARDS

- NSF ACCESS Advanced Computing Allocation (≈\$10,000 GPU resources)

2026

• Graduate College Professional Development Funds

2023

• Analytics Instructional Fee Award

2018

ACADEMIC SERVICES

- Workshop Organizer

◦ [1st Latent Visual Reasoning \(LVR\) Workshop](#), WACV 2027

• Conference Reviewer

◦ Annual Conference on Neural Information Processing Systems (NeurIPS), 2026

◦ ACM International Conference on Multimedia (ACM MM), 2026

◦ ACL Rolling Review (ARR), 2026

◦ International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2026

◦ IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

◦ Association for the Advancement of Artificial Intelligence (AAAI), 2025, 2026

◦ International Symposium on Biomedical Imaging (ISBI), 2025, 2026

◦ IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024–2026

• Journal Reviewer

◦ IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

◦ Nature - npj Digital Medicine

◦ IEEE Transactions on Multimedia (TMM)

◦ IEEE Transactions on Neural Networks and Learning Systems (TNNLS)